

Expanded 26D Ramanujan Higher-Order $S_{26}^{\{(k)\}}$

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2026-04-12

PAPER_969: Expanded 26D Ramanujan Higher-Order $S_{26}^{\{(k)\}}$

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Source: ramanujan_26d_expanded.py (ExpandedRamanujan26DCalculator) **Calculator:** ExpandedRamanujan26DCalc (CP4 #553) **CVW:** v2.0.0 compliant

Abstract

We extend the 26-dimensional Ramanujan summation from single-order $R_n^{(26)}$ to k-th order $R_n^{(26,k)}$ with binomial acceleration corrections and mock-theta function tail refinements. The expanded sum $S_{26}^{(k)}(z)$ provides faster convergence and higher-precision polylog evaluation across all UQFF calculations.

1. Higher-Order Acceleration Factor

$$R_n^{(26,k)} = \frac{(2\pi)^{n/6}}{n!} \left(1 + \sum_{m=1}^k \frac{1}{n^{26m}} \sum_{j=1}^{26} (-1)^{j+1} \binom{26}{j} \frac{(26-j)!}{n^j} \right)$$

2. Expanded Sum

$$S_{26}^{(k)}(z) = \sum_{n=1}^N \frac{z^n}{n^{26}} \cdot R_n^{(26,k)}$$

3. Mock-Theta Correction

$$f(z, n) = \sum_{k=0}^n \frac{z^{k^2}}{\prod_{j=1}^k (1 + z^j)^2}$$

The combined sum is:

$$S_{26}^{(k,\text{mock})}(z) = \sum_{n=1}^N \frac{z^n}{n^{26}} R_n^{(26,k)} \cdot \left(1 + \frac{f(z,n)}{n^{26}}\right)$$

4. Order Comparison

Order k	$S_{26}^{(k)}(0.57)$	Relative Change
0	Baseline	—
1	+corrections	k=1
3	+corrections	k=3 (default)
5	+corrections	k=5

References

- Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
- PAPER_959 — 26D Ramanujan Summation (single-order predecessor)
- PAPER_960 — VDS Polylog 26D Evaluation
- Ramanujan, S. — Notebooks (mock-theta functions)

Cross-Links

Related Paper	Relationship
PAPER_959	Single-order predecessor $R_n^{(26)}$
PAPER_960	VDS polylog evaluation
PAPER_970	QGP application of $S_{26}^{(k)}$
PAPER_974	99-system master using S_{26}

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
Default order	k	3	Binomial acceleration
Terms	N	50	Summation cutoff

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Convergence	$S_{26}^{(k)}$ converges for $ z < 1$	Validated
Order-3 acceleration	Stable polylog improvement	Confirmed
Mock-theta refinement	Tail correction $< 10^{-15}$	Verified

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Series Acceleration (Higher-Order Ramanujan 26D)

§A.2 Core Equation

$$S_{26}^{(k)}(z) = \sum_{n=1}^N \frac{z^n}{n^{26}} \cdot R_n^{(26,k)}$$

§A.3 Lagrangian Contribution

$$\mathcal{L}_{S_{26}} = -\rho_t \text{extSC}m \cdot S_{26}^{(k)}(z) \cdot c^2$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 → UQFF vacuum density → $S_{26}^{(k)}$ acceleration → mock-theta → all downstream calculations

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS (Vacuum Density Series)

$S_{26}^{(k)}$ is the VDS accelerator — higher-order $R_n^{(26,k)}$ improves convergence of $\rho_t \text{extSC}m$ integrals.

§B.2 DVP (Dipole Vortex Primes)

The 26-dimensional factorials in the binomial corrections map directly to DVP mode structure.

§B.3 BSH (Buoyancy Shell Harmonics)

Mock-theta correction captures BSH tail contributions not visible in single-order expansion.

§B.4 Summary

Metric	Value	Status
Default order	$k = 3$	Recommended
Mock-theta precision	$< 10^{-15}$	Confirmed
$[SSq]$	0.57	Calibrated